Make Compound Sentences Simple to Analyze: Learning to Split Sentences for Aspect-based Sentiment Analysis

Anonymous ACL submission

Abstract

In the domain of Aspect-Based Sentiment Analysis (ABSA), which has notably progressed alongside the evolution of generative models. Despite these advancements, ABSA remains a vibrant area of research due to the challenges associated with fully capturing the nuances of human emotions. Accurate prediction and sentiment extraction, especially the extraction of sentiment quadruplets, continue to pose significant challenges. The extraction of quadruplets in sentiment analysis tasks face difficulties as sentences grow more complex, with each containing quad elements. To address this, our focus shifts towards simplifying sentence structures to facilitate easier recognition of these elements, while also crafting a model that seamlessly integrates with various ABSA tasks. We introduce the Aspect Term Oriented Sentence Splitter (ATOSS), which simplifies complex compound sentences into clearer, simpler forms, thus clarifying their structure and intent. This approach not only preserves the essential meaning and structure of the sentences but also significantly enhances the extraction process, outperforming existing methods in both ASQP and ACOS tasks, which are the primary tasks for extracting quadruplets.

1 Introduction

Aspect-based sentiment analysis (ABSA) is a crucial task for understanding nuanced opinions and sentiments at a granular level. This technique identifies specific aspects of entities and evaluates their associated sentiments, providing richer contextual insights. Recently, ABSA has progressed beyond simple sentiment classification to tackle more complex structures like sentiment triplets and quadruplets, which include aspect term, opinion term, sentiment polarity, and aspect category for quadruplets. Efforts such as Aspect Sentiment Quad Prediction (ASQP) and Aspect-Category-Opinion-Sentiment (ACOS) reflect this shift towards predicting comprehensive sentiment tuples.

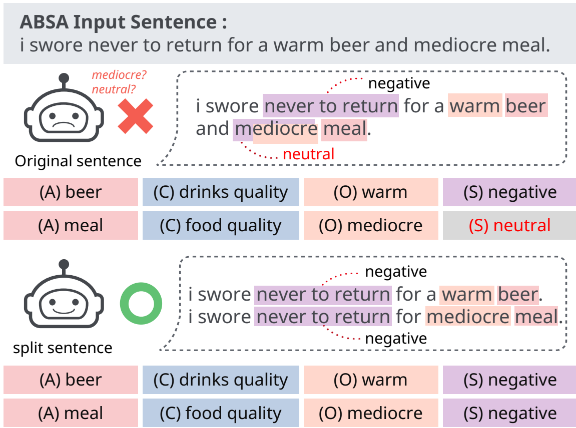

Figure 1: Existing models fail to predict quads in sentences with compound syntactic structures but succeed when the sentences are split into simpler structures.

Recently, generative approaches have shown promising performance in ABSA. Zhang et al. (2021b) and Gou et al. (2023) propose a framework that uses a sequence-to-sequence learning to transform an input sentence into predetermined output formats for predicting sentiment quadruplets. On the other hand, there has been an attempt to solve the task by utilizing large language models (LLMs) as zero-shot ABSA models, employing manually crafted zero-shot or few-shot prompts. The use of generative approach, whether based on fine-tuned model or LLM prompting, has significantly advanced the field, pushing towards more accurate and comprehensive sentiment analysis.

Despite state-of-the-art performances, recent ABSA models still suffer from the ambiguity of aspect-level sentiments in longer sentences. From our preliminary analyses, we observe that compound sentences are more prone to errors than simple ones, often due to multiple subjects with different states or context-dependent changes in a single subject. For example, in Figure 1 upper, the sentence "*I swore never to return for a warm beer and mediocre meal*" conveys a negative sentiment with "*never to return*". However, the opinion word "*mediocre*" describing the "*meal*" adds confusion to the overall sentiment of the sentence. By splitting this sentence into "*I swore never to return for a warm beer*" and "*I swore never to return for a mediocre meal*", we can clearly identify the intended ACOS quads, as depicted in Figure 1 lower. Additionally, sentences with lists, conjunctions, causal relationships, or storytelling elements are difficult to interpret clearly, requiring complex reasoning and judgment similar to human thought processes. Our research aims to enable models to easily find intent within simple and clear sentence structures, avoiding confusion in less complex and intertwined sentence constructions.

Motivated by the observations, we present a *plug-and-play* module, named **A**spect **T**erm **O**riented **S**entence **S**plitter (ATOSS), which helps to accurately identify quads by transforming an original, complicated sentences into simple and easy ones for ABSA. Through this text transformation, an input sentence is split to reduce ambiguity and guide the precise and clear prediction of quads. Specifically, our ATOSS module is optimized via LLM distillation and preference alignment (Figure 3). We first obtain split sentences by prompting LLMs and train our module to generate the split sentence given an original sentence. Additionally, we address any ambiguities or splitting biases by further tailoring our module for a target ABSA model; to this end, we adopt preference alignment (Rafailov et al., 2023) with sentence pairs of preferred-dispreferred splitting results. By doing so, ATOSS is fine-tuned to be further helpful for enhancing a target ABSA model's accuracy in quad prediction.

Our extensive experiments on four popular benchmark datasets demonstrate that ATOSS significantly enhances the prediction accuracy of state-of-the-art ABSA models, including fine-tuned models and LLM prompting. Specifically, ATOSS effectively reduces the error rates associated with incorrectly predicting aspect terms by facilitating the identification of aspect terms in input sentences for each sentiment quad. Moreover, our module can be seamlessly integrated into the inference pipeline for other ABSA tasks, highlighting its high level of generalizability. For reproducibility, our codes are publicly available.[1]

[1]https://anonymous.4open.science/r/ATOSS-478B/README.md

The main contributions of our paper can be summarized as follows:

- We propose a plug-and-play module that splits a compound sentence into simple sentences, which is integrated into existing ABSA models without updates of their parameters.

- ATOSS is optimized to align with the sentence preference, which is helpful to enhance target ABSA model's accuracy in quad prediction.

- Experiments show that ATOSS improves ABSA model's accuracy in quad prediction by effectively reducing aspect-level errors, with its great generalizability to other ABSA tasks.

2 Related Work

In this section, we review the literature on (1) state-of-the-art approaches to aspect-based sentiment analysis (ABSA), and (2) recent efforts to distillation of large language models' (LLMs) remarkable reasoning ability on various tasks into a small language models (LMs).

2.1 Sentiment Quadruplet Prediction

ABSA has been widely studied in recent years, focusing on extracting sentiment-related elements for more fine-grained sentiment analysis. Early studies focused on predicting single or dual sentiment elements, such as aspect term extraction (Ma et al., 2019) and aspect-level sentiment classification (Zhang and Qian, 2020). Later, more challenging ABSA tasks have been proposed to predict sentiment triplets or quadruplets, including Aspect Sentiment Triplet Extraction (ASTE) (Peng et al., 2020), Aspect Category Opinion Sentiment (ACOS) (Cai et al., 2021), and Aspect Sentiment Quad Prediciton (ASQP) (Zhang et al., 2021b).

Recently, most existing works have attempted to tackle quad prediction problems by using generative approaches. Zhang et al. (2021b) solved the ASQP task by transforming target quads into natural language sentences, utilizing the knowledge from the pre-trained generative model, thus leading to better performance. Hu et al. (2022) proposed methods to augment ASQP dataset given the order-free property of the quadruplet based on templates. Gou et al. (2023) introduce an element order-based prompt learning method that improves sentiment tuple prediction by aggregating multi-view results.

Task	Datasets	Ratio of S / C	Acc of S	Acc of C
ASQP	Rest15	32.93 / 67.07	53.57	40.08
	Rest16	32.63 / 67.37	57.87	50.21
ACOS	Laptop16	32.72 / 67.28	39.28	30.23
	Rest16	32.16 / 67.84	54.45	48.25

Table 1: Proportion of simple(S) / compound(C) inputs, and quad prediction accuracy (i.e., recall) of existing ABSA models for simple and compound sentences.

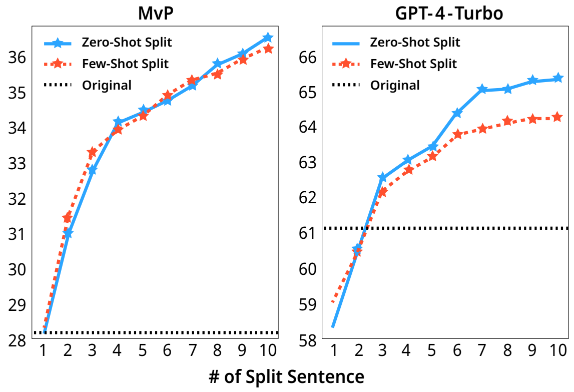

Figure 2: Performance changes of ABSA models (Left: MvP, Right: GPT-4-turbo) w.r.t. the number of candidate split sentences. (Task: ACOS, Dataset: Rest16)

Most recent studies have focused on enhancing the output and reasoning process of generative models. However, the complexity of thought in lengthy and intricate sentences still poses significant challenges for clear inference, emphasizing the need for more refined strategies capable of handling such complexity with greater precision.

2.2 Distillation of LLM's Reasoning Ability

Knowledge distillation, which trains smaller models based on larger models, aims to reduce their size and latency while maintaining accuracy and generalization capabilities (Hinton et al., 2015; Sanh et al., 2019). With the remarkable performance of LLMs across a wide range of tasks, extensive research has been conducted to leverage their capabilities for knowledge distillation. Recent work has focused on distilling the reasoning capabilities of LLMs to smaller LMs. Fine-tune-CoT (Ho et al., 2022) has demonstrated outstanding performance across various tasks by enabling LLMs to generate diverse reasoning paths and distill them into LMs. Some studies have conducted more detailed, task-specific knowledge distillation using LLMs (Magister et al., 2022; Chae et al., 2023; Hsieh et al., 2023). There has been a recent attempt to utilize reasoning in ASQP as well, aiming to solve ASQP by leveraging the reasoning ability distilled from LLMs (Kim et al., 2024).

3 Preliminaries

In this section, we formally define our target ABSA tasks and analyze ABSA models' behavior based on the structural complexity of input sentences.

3.1 Problem Formulation

In this work, we focus on tasks of predicting aspect-level sentiment from input sentences, in the form of structured quadruplets, i.e., ASQP and ACOS. Formally, given an input sentence, these task aim to predict all aspect sentiment quadruplets consisting of four components, i.e., $\{(\texttt{at}, \texttt{ot}, \texttt{ac}, \texttt{sp})\}$. The aspect term "at" and opinion term "ot" are detected within the sentence, while the aspect category "ac" and sentiment polarity "sp" are classified within their respective predefined sets.

The aspect term includes the possibility of being null, denoted as "NULL", to indicate cases where it is not explicitly mentioned. Note that ACOS differs from ASQP in its definition of the opinion term, focusing on more implicit aspects and opinions, which may result in the opinion term being null as well. The aspect category is classified as an element within the category set predefined for each domain or dataset; for example, in restaurant review datasets, it includes various categories that can group the domain-specific aspects, such as "*food prices*" and "*ambience general*". The sentiment polarity is predicted as one of the three sentiment classes: "*positive*", "*neutral*", and "*negative*", each indicating the corresponding emotional disposition conveyed. The further descriptions about the target tasks are provided in Appendix A.

3.2 Analysis of ABSA Performance based on Sentence Structural Complexity

We first investigate the prediction accuracy of existing ABSA models according to the degree of complexity in an input sentence structure. To this end, we categorize all test inputs into two sets: *simple sentence* and *compound sentence*. In this work, we define simple sentence as one that contains only a single independent clause, annotated with a single sentiment quad. In contrast, compound sentence is linked by conjunctions like "*and*", "*or*", "*but*", and punctuated with commas; it can be annotated with one or more sentiment quads.

Table 1 reports the error rates of state-of-the-

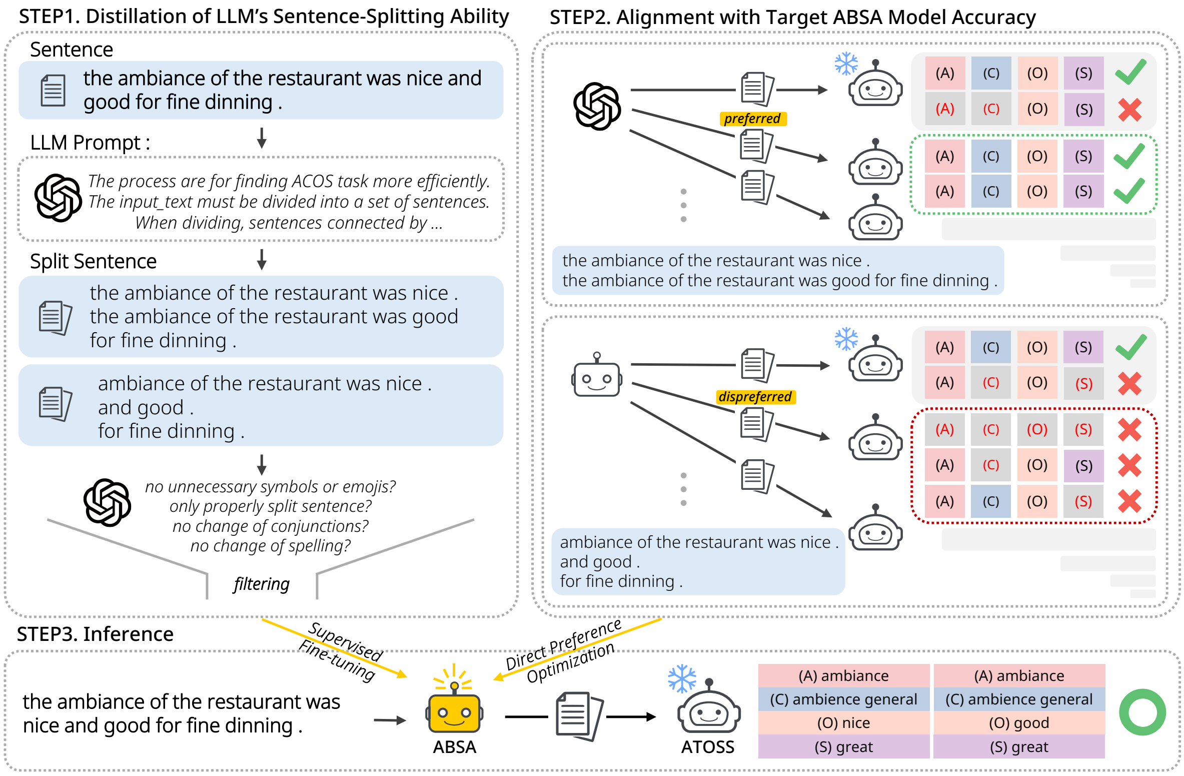

Figure 3: Overall framework for training and utilizing ATOSS for ABSA tasks. The training process involves in (1) distillation of LLM's capability for sentence splitting, and (2) alignment with a target model's prediction accuracy. The inference process predicts the quads by taking sentences split by ATOSS as the input. ATOSS is a *plug-and-play* module for enhancing the predication accuracy in that parameters of the target model are not updated.

art ABSA models for both simple sentence and compound sentence inputs. From the results, we observe a high error rate in compound sentence across all tasks. As illustrated in Figure 1, when multiple sentiment quads exist for a single aspect term, the sentence becomes longer and more complex. In such cases, the model struggles to detect the aspect term accurately, especially when the distance between the aspect term and the sentiment component in the text is relatively large, making accurate quad prediction difficult. A similar issue arises when multiple aspects each have multiple associated sentiment quads within the same sentence. Based on these observations, we hypothesize that splitting a compound sentence into simple ones based on aspect terms could reduce the error rate and lead to higher overall accuracy. We provide more detailed analysis in Section 5.2.

In addition, we conduct a preliminary study to validate our hypothesis that simplifying sentences via splitting can enhance their clarity and thereby improve ABSA models' quad prediction accuracy. To this end, we examine how much F1 score in quad prediction improves when using split sentences instead of the original ones. Specifically, for each test input, we generate 10 various split sentences by prompting LLMs to split sentences in both zero-shot and few-shot manners. We then select the best split sentence that yields the highest F1 accuracy (i.e., *oracle* voting). Figure 2 shows the changes in prediction F1 score as the number of split sentences increases.

Both the fine-tuned model (i.e., MvP) and the prompting-based LLM (i.e., GPT-4-turbo) demonstrate significant performance improvements with split sentences, especially when more candidates of split sentences are available. Notably, split sentences obtained through zero-shot prompts, which result in more diverse split forms compared to few-shot prompts, show great potential for enhancing the performance of existing ABSA models. Consequently, we confirm that the strategy of splitting sentences into appropriate simple forms can indeed aid in enhancing ABSA performance.

4 Methodology

In this section, we present a *plug-and-play* module that splits an input text into multiple simple sentences that facilitates ABSA tasks. Our proposed module, named Aspect Term Oriented Sentence Splitter (ATOSS), is a small LM trained via knowledge distillation from a teacher LLM and further refined to align its output with helpfulness for enhancing the target ABSA model's accuracy. The overview of ATOSS is illustrated in Figure 3.

4.1 Distillation of LLM's Splitting Ability

We first optimize our sentence splitter to generate diverse split sentences given an original input sentence. To train the sentence splitter, we augment the ABSA training dataset of sentence-quads pairs $(s, \mathcal{Q})$ into $\mathcal{X} = \{(s, s', \mathcal{Q})\}$, where s' is the split sentence from the original sentence s.

Split sentence generation As mentioned in Section 3.2, existing datasets contain both simple and compound sentences for ABSA models to process. For the simple sentences, the models can perform the tasks well without any additional steps, so we retain them without modification. However, the compound sentences often have a complex structure, such as multiple quads oriented from a single aspect term, making it challenging for existing models to perform the tasks. To address this issue, we only split these compound sentences.

To distill a teacher LLM's sentence splitting ability into ATOSS, we generate training data for distillation by prompting the LLM. In this step, we simply adopt few-shot prompting for split sentence generation, allowing the LLM to effectively split s into s' via in-context learning of given examplers. By providing the LLM with $(s, \mathcal{Q})$, we guide it to split s based on its ground-truth aspect terms of $\mathcal{Q}$. To explore effective and appropriate s', we prompt the LLM to generate 10 diverse s' for each s.

Split sentence selection However, as s' may still contain noise, we introduce an additional filtering process to select K split sentences that best meet the specific criteria, within the set of generated s'. This filtering process is also performed via LLM prompting with manually written splitting criteria. In the end, we construct the dataset for fine-tuning the ATOSS. The prompt can be found in the Appendix B.

Supervised fine-tuning (SFT) We train the model to predict the target s' given an input s. With the input-target pair (s, s'), we fine-tune the sequence-to-sequence LM by minimizing the following negative log-likelihood loss:

$$\mathcal{L}_{NLL} = -\log p(s'|s) = -\sum_{t=1}^{T} \log p(s'_t|s, s'_{<t}),$$

where T is the length of the target sequence s and $s'_{<t}$ denotes previously generated tokens.

Note that the purpose of this step is to obtain a general sentence splitter by distilling an LLM's sentence splitting ability; therefore, we use all (s, s') samples collected from multiple available datasets.

4.2 Alignment with Sentence Preference

Even though the general splitter is able to split the sentence to some extent, but the general split sentence may not always be an optimally processed input for a specific model. To further refine the splitter for a target dataset, task, and ABSA model, we additionally tune ATOSS based on preference alignment strategy. While shorter sentences are generally easier for tasks than longer ones, the optimal format for split sentences may vary depending on the models and datasets. To address this issue, we apply Direct preference optimization (DPO) (Rafailov et al., 2023) to the ATOSS for each specific model and dataset to perform the tasks.

Preferred sentence selection We obtain s' that is optimal based on manual criteria, but this may exclude other optimal s' formats that we do not consider. To mitigate this issue, we generate 10 diverse s' through zero-shot prompting with the LLM and measure the sentence-level F1 score from the ABSA model's inference by providing s and s' as input. We use this comparison to select the preferred sentences: if the s has a higher F1 score, no preferred sentence is chosen; if the scores are equal, s is retained if it is simple, otherwise, we select s' with the most conjunctions; if s has a lower F1 score, we select all distinct s' with higher F1 scores.

Dispreferred sentence selection Even we constuct the appropriate dataset, the model may fail to generate optimal s' for ABSA model. To alleviate this issue, we generate s' utilizing ATOSS in the same manner as in the preferred sentence selection. We use this comparison to select the dispreferred sentences: if the original sentence has a lower F1 score, no dispreferred sentence is chosen; if the scores are equal, we select s' with the lowest similarity to s if s is simple, or the highest similarity if s is complex; if s has a higher F1 score, we select

s' with the highest similarity to s among those with lower F1 scores.

Direct preference optimization Through the aforementioned process, we can construct preference pairs $P = \{(s, p^+, p^-)\}$ using the preferred sentence p^+ and the dispreferred sentence p^-. We apply DPO on our sentence splitter θ to train a preference-tuned sentence splitter θ^* that minimizes the following objective:

$$L_{\text{DPO}}(\theta^*; \theta) = -\mathbb{E}_{(s,p^+,p^-)\sim P} \log \sigma \left[r(s, p^+) - r(s, p^-) \right],$$

where $r(s,p) = \frac{p_{\theta^*}(p|s)}{p_\theta(p|s)}$. By optimizing the model using preferred-dispreferred sentence pairs, our obtained model θ^* is trained to prefer sentences that are clearly split based on the aspect term while avoiding those that are ambiguously or unclearly split. Note that θ has been specifically trained for each ABSA model, since the preferred split sentences vary across different models.

4.3 ATOSS-Augmented Quad Prediction

During inference, our final ATOSS model obtained via two-step optimization transforms an input sentence into the split one, which is fed into the ABSA model. This stage adopts ATOSS tailored for a specific ABSA model and dataset. This plug-and-play module allows existing ABSA models to improve performance by processing input sentences optimized for each task without updates of model parameters. In other words, as ABSA models themselves are not required any tuning, ATOSS can be universally applied to any off-the-shelf ABSA models, or to any ABSA tasks focusing on aspect-level sentiments. Note that it can also be adopted to proprietary LLMs, such as GPT-3.5-turbo and GPT-4-turbo, which cannot be tuned, demonstrating excellent applicability and flexibility.

5 Experiments

We conducted our experiments while addressing the following research questions:

- **RQ1:** Can ATOSS enhance the quad prediction accuracy of existing ABSA models?
- **RQ2:** Can ATOSS reduce aspect-level errors in quad prediction for existing ABSA models?
- **RQ3:** Can ATOSS be effective for other ABSA tasks beyond quad prediction?

5.1 Experimental Settings

Tasks and datasets We validate the effectiveness of ATOSS on 4 datasets across 2 tasks, ASQP and ACOS. For ASQP, we utilize two restaurant domain datasets, i.e., Rest15 and Rest16 (Pontiki et al., 2015, 2016). In the case of ACOS, we adopt restaurant-ACOS and Laptop-ACOS datasets, i.e., Rest16 and Laptop16 (Cai et al., 2021). Refer to Appendix C for more details.

Implementation details We employ the T5-base model (Raffel et al., 2020) from Huggingface Transformers[2] (Wolf et al., 2020) as the backbone model of our splitter. We adopt the *plug-and-play* approach in our experiments, maintaining the existing parameters of each ABSA model while only tuning the parameters of the ATOSS model. To filter out noisy sentences and select those that best meet the criteria, we set K=2. We use two variants of our model: the splitter trained by using only LLM distillation, named *General* (Section 4.1), and the one aligned for preference, named *Specific* (Section 4.2). Refer to Appendix D for more details.

Evaluation metrics For all tasks, a sentiment tuple is considered correct if and only if every element matches the gold tuple exactly. We utilize F1 scores as the primary evaluation metric (Zhang et al., 2021a; Mao et al., 2022), with all reported F1 scores are randomly selected seed. We used the same F1 score metric to evaluate GPT-4-turbo in a single run. We also reported the precision (Pre) and recall (Rec) scores.

ABSA models For ABSA models, we use generative method-based models that have recently shown outstanding performance, i.e., **Paraphrase** (Zhang et al., 2021b) and **MvP** (Gou et al., 2023). We also use the LLMs without fine-tuning, employing it through zero-shot prompting. For the LLMs, we use **GPT-3.5-turbo** (`gpt-3.5-turbo-0125`) and **GPT-4-turbo** (`gpt-4-1106-preview`).[3]

5.2 Effectiveness of ATOSS (RQ1 & RQ2)

5.2.1 Performance Comparison

Table 2 presents the ASQP and ACOS performance of various models. Overall, the ABSA models integrating our ATOSS module, which takes split sentences as inputs, leads to better performance compared to the ones that use original sentences as

[2] https://github.com/huggingface/transformers

[3] https://chat.openai.com/

Methods	ASQP						ACOS					
	Rest15			Rest16			Laptop16			Rest16		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
Paraphrase (Zhang et al., 2021b)	43.68	47.42	45.48	56.13	59.57	57.80	42.61	42.98	42.80	61.61	57.45	59.46
+ ATOSS (General)	45.44	47.04	**46.23**	57.28	59.57	**58.40**	42.44	42.29	42.36	61.43	57.02	59.14
+ ATOSS (Specific)	45.32	46.92	46.11	57.21	59.57	58.37	43.00	43.93	**43.46**	64.03	59.01	**61.41**
MvP (Gou et al., 2023)	49.81	48.68	49.24	61.33	62.33	61.82	43.76	43.69	43.72	61.27	57.86	59.52
+ ATOSS (General)	51.99	49.18	50.55	61.61	61.45	61.53	45.21	42.91	44.03	61.05	56.99	58.95
+ ATOSS (Specific)	51.99	49.18	**50.55**	62.80	61.70	**62.25**	45.32	43.17	**44.22**	63.12	58.30	**60.61**
GPT-3.5-turbo	14.49	15.22	14.85	20.99	22.28	21.62	8.12	8.60	8.35	24.00	23.58	23.79
+ ATOSS (General)	15.97	17.74	**16.81**	21.15	23.53	22.27	8.37	8.79	**8.58**	25.80	26.31	**26.05**
+ ATOSS (Specific)	15.77	17.61	16.64	21.73	23.90	**22.77**	7.94	8.44	8.18	25.40	25.87	25.64
GPT-4-turbo	20.11	25.96	22.66	23.76	28.16	25.77	9.24	10.51	9.83	28.44	30.68	28.99
+ ATOSS (General)	21.11	25.91	23.26	23.52	28.79	25.89	9.31	10.77	**9.98**	28.18	31.66	29.82
+ ATOSS (Specific)	21.59	26.67	**23.86**	25.73	30.91	**28.08**	9.16	10.85	9.94	28.68	31.88	**30.20**

Table 2: Performance (%) of various ABSA models and the ones equipped with our text splitter.

inputs. This improvement is consistently observed across fine-tuned ABSA models and when applied to LLM prompting; this highlights the efficacy of our sentence splitting approach in enhancing the accuracy of existing ABSA models without parameter tuning and extensive modifications. Both the general and specific versions of ATOSS yield improved results, demonstrating its consistent ability to generate clear split sentences for ABSA models. Moreover, it effectively tailors split sentences to match the specific structural patterns easily captured by each model.

Methods	Rest15			Rest16		
	Pre	Rec	F1	Pre	Rec	F1
MvP	66.20	60.95	63.46	72.93	69.62	71.23
+ ATOSS (G)	66.45	60.47	63.32	72.44	68.22	70.26
+ ATOSS (S)	67.14	61.18	**64.02**	73.40	69.38	**71.33**

Table 3: Performance (%) of various ABSA models and the ones equipped with ATOSS in the *cross-task* setting. ATOSS is trained for ASQP then tested for TASD.

5.2.2 Aspect-Level Error Analysis

In the Table 4, we evaluate the aspect-level F1 score. We observe that splitting into simpler sentences improves the detection of aspect terms. To validate our approach, we initially employed oracle voting and conducted experiments. We benchmarked the original scores of the ABSA task's test set, simplifying sentences and selectively splitting compound sentences to optimize quad prediction. Particularly challenging were cases where hard to detect aspect term and other sentiment elements annotated with one quadruplet or complex structural issues obscured relationships among aspects. Our approach focused on guiding sentence splitting to maintain coherence and context, specifically aligning with aspects for proper continuity. Following Supervised Fine-Tuning (SFT), we observed enhanced scores for compound sentences. In Table 5, domain-specific preference alignment led to scores surpassing the baseline.

This outcome underscores the significance of focusing on aspects to improve relationships across categories, opinions, and sentiments, as evidenced by our achieved scores.

Methods	ACOS		ASQP	
	Laptop16	Rest16	Rest15	Rest16
Paraphrase	**83.26**	80.81	74.04	**80.84**
+ATOSS	83.24	**82.20**	**74.12**	80.58
MvP	84.02	83.63	76.35	82.54
+ATOSS	**85.05**	**83.64**	**77.62**	**83.38**
GPT-4-turbo	66.62	62.83	61.83	**65.35**
+ATOSS	**68.84**	**63.36**	**62.81**	64.62

Table 4: Performance (%) of various ABSA model in terms of aspect-level F1.

5.3 Generalizability of ATOSS in ABSA (RQ3)

We evaluate our approach in a cross-task setting by utilizing the splitter trained for the quad prediction task, to preprocess the input of a fine-tuned model performing TASD. When evaluating the performance, we integrate a general splitter trained on quad prediction and the specific splitter tailored to the restaurant dataset. In Table 3, we integrate ATOSS with generative model and LLMs and subsequently evaluated their performance. The results

Model	Task	Dataset	Single Sentence						Compound Sentence					
			Base Model			+ ATOSS (S)			Base Model			+ ATOSS (S)		
			Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
Paraphrase	ASQP	Rest15	55.74	57.30	**56.51**	55.87	57.14	56.50	44.93	40.15	42.41	44.32	42.13	**43.20**
		Rest16	64.17	67.42	65.75	65.93	68.57	**67.23**	58.17	53.13	55.54	57.70	54.19	**55.89**
	ACOS	laptop16	48.94	51.48	50.18	49.12	51.87	**50.45**	41.05	39.96	40.49	42.26	40.41	**41.31**
		Rest16	61.54	65.98	**63.68**	61.24	65.64	63.37	56.24	60.31	58.20	58.33	63.54	**60.82**
MvP	ASQP	Rest15	60.89	61.58	61.24	60.44	62.15	**61.28**	46.49	44.98	45.72	49.30	45.47	**47.31**
		Rest16	70.27	73.03	**71.63**	69.73	72.47	71.07	58.69	59.26	58.97	60.67	58.62	**59.62**
	ACOS	laptop16	51.16	52.06	51.58	51.29	52.06	**51.67**	41.50	41.17	41.33	43.37	40.49	**41.88**
		Rest16	67.20	69.02	68.10	67.55	69.02	**68.28**	55.62	55.06	55.24	58.21	56.69	**57.44**
GPT-4-turbo	ASQP	Rest15	38.14	41.81	39.89	38.78	42.94	**40.75**	28.59	35.11	**31.52**	25.95	33.01	29.06
		Rest16	30.41	33.15	31.72	33.85	36.52	**35.14**	31.34	38.00	34.35	32.68	40.42	**36.14**
	ACOS	laptop16	13.45	13.96	**13.70**	13.31	13.96	13.63	8.22	9.60	8.86	8.39	10.27	**9.23**
		Rest16	26.15	27.72	**26.91**	23.86	25.54	24.67	28.37	30.74	29.51	29.35	32.92	**31.04**

Table 5: Performance (%) of various ABSA models for the quad prediction task in terms of sentence-level F1 scores.

consistently demonstrate improved performance across all scenarios, highlighting the effectiveness of simplyfying the sentence through splitting in cross-task settings.

6 Conclusion

In this paper, we aim to simplify sentences through strategic sentence splitting, thereby improving the performance of quadruplet prediction in ABSA tasks. This process retains essential elements while dividing sentences, thereby making them more concise and clear. Our approach involves initially generating split sentences through prompting, allowing us to select clear sentences and discard noisy and ambiguous ones, thereby upgrading the model's performance by providing inputs that are easier for the model to process. Based on these points, we demonstrate that ATOSS, as a play-and-plug module, can be seamlessly integrated with various fine-tuned ABSA models as well as untunable LLMs, improving performance without requiring any modifications to the models themselves. Our research highlights the significant impact of sentence structure on sentiment analysis, presenting substantial implications for the field of ABSA.

7 Limitations

Despite achieving state-of-the-art performance, our study encounters several limitations. First, the available dataset was insufficiently large, limiting our ability to train effectively. The scarcity of ground truth data containing all four quadruplets necessitated the use of cross-training with unseen data, which might have yielded better results with a more extensive dataset. Additionally, the relatively small size of our initial model suggests that scaling up the model could potentially improve outcomes. Secondly, the cost associated with our methodology presents another significant challenge. The use of the GPT-3.5-turbo and GPT-4-turbo entails substantial expenses, particularly due to the repeated prompting required during the data regeneration process for sentence splitting. Each instance of data creation incurs costs, impacting the scalability of our approach. Lastly, unlike LLM, the dependency of our plug-and-play model on underlying ABSA models presents a constraint. Since our model is not independently operational but relies on existing ABSA models, there can be variability in performance based on the quality and characteristics of the ABSA models employed.

8 Ethical Statement

We employ datasets that are well-recognized and previously utilized within the scientific community, ensuring transparency and integrity in our experiments. Our methodologies and findings do not cause harm to any individuals or groups.

We are aware of the potential biases in sentiment polarity predictions that may arise from using large pre-trained language models, as these models can reflect societal biases present in their training data (Tan and Celis, 2019). We recognize the importance of ongoing efforts to address these biases. Moreover, we emphasize the necessity of continuous monitoring and evaluation to prevent our smaller downstream models from replicating or amplifying the biases of their larger language model counterparts.

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A Problem Formulation Details

The aspect term includes denotes cases where it is not explicitly mentioned, which is represented by“NULL”.

The aspect category includes that “food prices”, “food style options”, “service general”, “drinks prices”, “ambience general”, “drinks quality”, “location general”, “restaurant prices”, “restaurant general”, “drinks style options”, “food general”, “restaurant miscellaneous”, “food quality”.

The sentiment polarity of ASQP is predicted as one of the three sentiment classes: “*positive*”, “*neutral*”, and “*negative*”, each indicating the corresponding emotional disposition conveyed.

Mapping in ACOS, for example, involves mapping the sentiment polarity ’POS’ to ’great’ and the ’NULL’ label of the opinion term ’o’ to ’it,’ following the methodology outlined in previous work by Zhang et al. (2021a)."

Note that ACOS differs from ASQP in defining the opinion term, as it focuses on more implicit aspects and opinions.

B Detailed Prompts for Sentence Splitting

To generate split sentences, we provided the detailed prompts used for asking LLM to split an input sentence into a set of simple sentences. Tables 6 and 7 zero shot prompt for Split sentence generation few shot prompt for Direct Preference Optimization

C Dataset Statistics

Tables 8 shows the data statistics of all datasets of the ASQP, ACOS and TASD task.

D Implement Details

For SFT training, The batch size for T5-based models is set at 64. The validation batch size is set at 8. Training consists of 50 epochs with a learning rate of 6e-5. Early stopping is implemented with a patience of 20 epochs.

For DPO training, The batch size for T5-based models is set at 8. The validation batch size remains at 8. Training is conducted over 1 epoch with a learning rate of 1e-4. Early stopping is also set with a patience of 20 epochs. The beta parameter is set at 0.1 (default), and the loss function used is sigmoid.

Zero Shot Prompt of ACOS rest16
[Task Description] The process are for finding ACOS task more efficiently. The input_text must be divided into a set of sentences. When dividing, sentences connected by conjunctions must be divided into individual sentences with it's conjunctions. Divide the sentences in a way that is more advantageous for the ACOS task to better identify the quads. This process must specify the subject in every sentence. This process must retain the existing spellings exactly as in the input_text. This process must also retain the existing spacings exactly as in the input_text. This process must lowercase the alphabet if it's capitalized. If the sentence is too short to divide or does not need to be divided, use the input_text sentence as is. No numbering, line breaks, "output_text:" or explanations are needed.

Table 6: The zero-shot prompt for splitting an input sentence on the ABSA task.

Few Shot Prompt of ACOS rest16
[Task Description] The process are for finding ACOS task more efficiently. The input_text must be divided into a set of sentences. When dividing, sentences connected by conjunctions must be divided into individual sentences with it's conjunctions. Divide the sentences in a way that is more advantageous for the ACOS task to better identify the quads. This process must specify the subject in every sentence. This process must retain the existing spellings exactly as in the input_text. This process must also retain the existing spacings exactly as in the input_text. This process must lowercase the alphabet if it's capitalized. If the sentence is too short to divide or does not need to be divided, use the input_text sentence as is. No numbering, line breaks, "output_text:" or explanations are needed. **[Example 1]** **input_text:** maybe it was the great company (i had friends visiting from philly – yes , it was not a date this time) or the super reasonable price point , but i just can ' t say enough good things about this brasserie . [['brasserie', 'restaurant general', 'positive', 'good'], ['brasserie', 'restaurant prices', 'positive', 'good']] **output_text:** maybe it was the great company . i had friends visiting from philly . yes , it was not a date this time . or the super reasonable price point . but i just can ' t say enough good things about this brasserie . [['brasserie', 'restaurant general', 'positive', 'good'], ['brasserie', 'restaurant prices', 'positive', 'good']] **[Example 2]** **input_text:** i had their eggs benedict for brunch , which were the worst in my entire life , i tried removing the hollondaise sauce completely that was how failed it was . [['eggs benedict', 'food quality', 'negative', 'worst']] **output_text:** i had their eggs benedict for brunch . which were the worst in my entire life . i tried removing the hollondaise sauce completely that was how failed it was . [['eggs benedict', 'food quality', 'negative', 'worst']] **[Example 3]**...

Table 7: The few-shot prompt for splitting an input sentence on the ABSA task (Examples 3 to 10 are omitted in this table).

Few-Shot Prompts for SFT Filtering
[Task Description] Given the following input sentence, select best 2 output sentence number from the 10 provided that satisfy the following cases. No need of explanation. Give me just answer number. Best sentence: sentence that satisfy most of the following cases. Input: do n ’ t judge this place prima facie , you have to try it to believe it , a home away from home for the literate heart . no unnecessary symbols or emojis? no change of conjunctions? no change of spelling? only properly split sentence? Output: 1. do n ’ t judge this place prima facie. you have to try it to believe it. a home away from home for the literate heart. 2. do n’t judge this place prima facie. you have to try it to believe it. it is a home away from home for the literate heart. 3. do n ’ t judge this place prima facie. you have to try it to believe it. a home away from home for the literate heart. 4. don’t judge this place prima facie. you have to try it to believe it. it is a home away from home for the literate heart. 5. do n ’ t judge this place prima faci e. you have to try it to believe it. it ’ s a home away from home for the literate heart. 6. do n ’ t judge this place prima facie. you have to try it to believe it. it ’s a home away from home for the literate heart. 7. do n ’ t judge this place prima facie. you have to try it to believe it. a home away from home for the literate the heart. 8. do n ’ t judge this place prima facie. you have to try it to believe it. it is a home away from home for the literate heart. 9. do n’t judge this place prima facie, you have to try it to believe it, and a home away from home for the literate heart. 10. do n ’ t judge this place prima facie. you have to try it to believe it. a home away from home for the literate heart.

Table 8: The Few-Shot Prompts for SFT Filtering (Examples are omitted in this table).

Task	Dataset Cat#	**Train (POS/NEU/NEG)**	**Dev (POS/NEU/NEG)**	**Test (POS/NEU/NEG)**
ASQP	Rest15 #13	834 1,005/34/315	209 252/14/81	537 453/37/305
	Rest16 #13	1,264 1,369/62/558	316 341/23/143	544 584/40/177
ACOS	Rest16 #13	1,530 1,656/95/733	171 180/12/69	583 668/44/205
	Laptop16 #121	2,934 2,583/227/1,364	326 279/24/137	816 716/65/380
TASD	Rest15 #13	1,120 1,198/53/403	10 6/0/7	582 454/45/346
	Rest16 #13	1,708 1,657/101/749	29 23/1/297	587 611/44/204

Table 9: Dataset statistics for various tasks. #Cat refers to the number of aspect categories in the set. POS, NEU, and NEG denote the number of positive, neutral, and negative quads or triplets respectively.